8. Groups of Permutations

1. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 3 & 6 & 5 & 4 \end{pmatrix}$ **2.** $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 1 & 5 & 6 & 3 \end{pmatrix}$ **3.** $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 1 & 6 & 2 & 5 \end{pmatrix}$

4. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 1 & 6 & 2 & 4 & 3 \end{pmatrix}$ **5.** $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 6 & 1 & 5 & 4 & 3 \end{pmatrix}$

6. Starting with 1 and applying σ repeatedly, we see that σ takes 1 to 3 to 4 to 5 to 6 to 2 to 1, so σ^6 is the smallest possible power of σ that is the identity permutation. It is easily checked that σ^6 carries 2, 3, 4, 5 and 6 to themselves also, so σ^6 is indeed the identity and $|\langle\sigma\rangle| = 6$.

7. $\tau^2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 3 & 2 & 1 & 5 & 6 \end{pmatrix}$ and it is clear that $(\tau^2)^2$ is the identity. Thus we have $|\langle\tau^2\rangle| = 2$.

8. Because σ^6 is the identity permutation (see Exercise 6), we have

$$\sigma^{100} = (\sigma^6)^{16}\sigma^4 = \sigma^4 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 2 & 1 & 3 & 4 \end{pmatrix}.$$

9. We find that μ^2 is the identity permutation, so $\mu^{100} = (\mu^2)^{50}$ is also the identity permutation.

10. $\{\mathbb{Z}, 17\mathbb{Z}, 3\mathbb{Z}, \langle\pi\rangle\}$ is a subcollection of isomorphic groups, as are $\{\mathbb{Z}_6, G\}, \{\mathbb{Z}_2, S_2\}, \{S_6\}, \{\mathbb{Q}\}, \{\mathbb{R}, \mathbb{R}^+\}$, $\{\mathbb{R}^*\}$, $\{\mathbb{Q}^*\}$, and $\{\mathbb{C}^*\}$.

11. $\{1, 2, 3, 4, 5, 6\}$ **12.** $\{1, 2, 3, 4\}$ **13.** $\{1, 5\}$

14. We see that ϵ, ρ, and ρ^2 give the three positions of the triangle in Fig. 8.9 obtained by rotations. The permutations $\phi, \rho\phi$, and $\rho^2\phi$ amount geometrically to turning the triangle over (ϕ) and then rotating it to obtain the other three positions.

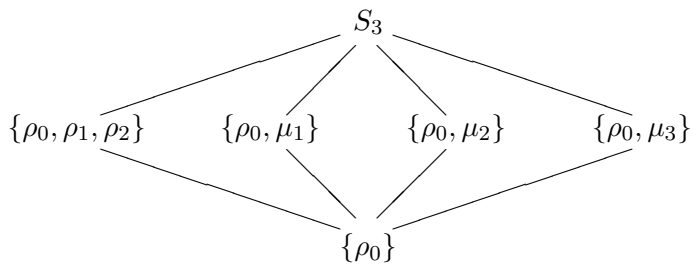
15. A similar labeling for D_4 is $\epsilon, \rho, \rho^2, \rho^3, \phi, \rho\phi, \rho^2\phi, \rho^3\phi$ where their ϕ is our μ_1. They correspond to our elements in the order $\rho_0, \rho_1, \rho_2, \rho_3, \mu_1, \delta_1, \mu_2, \delta_2$.

16. σ may have the action of any of the six possible permutations of the set $\{1, 2, 4\}$, so there are six possibilities for σ.

17. There are 4 possibilities for $\sigma(1)$, then 3 possibilities for $\sigma(3)$, then 2 possibilities for $\sigma(4)$, and then 1 possibility for $\sigma(5)$, for $4 \cdot 3 \cdot 2 \cdot 1 = 24$ possibilities in all.

18. **a.** $\langle\rho_1\rangle = \langle\rho_2\rangle = \{\rho_0, \rho_1, \rho_2\}$ and $\langle\mu_1\rangle = \{\rho_0, \mu_1\}$.

b.



19. (See the answer in the text.)

20. This group is not isomorphic to S_3 because it is abelian and S_3 is nonabelian. It is isomorphic to $\mathbb{Z}_6$.

	ρ^0	ρ	ρ^2	ρ^3	ρ^4	ρ^5
ρ^0	ρ^0	ρ	ρ^2	ρ^3	ρ^4	ρ^5
ρ	ρ	ρ^2	ρ^3	ρ^4	ρ^5	ρ^0
ρ^2	ρ^2	ρ^3	ρ^4	ρ^5	ρ^0	ρ
ρ^3	ρ^3	ρ^4	ρ^5	ρ^0	ρ	ρ^2
ρ^4	ρ^4	ρ^5	ρ^0	ρ	ρ^2	ρ^3
ρ^5	ρ^5	ρ^0	ρ	ρ^2	ρ^3	ρ^4

21. (See the answer in the text.)

22. We list matrices in order corresponding to the permutations $\rho_0, \rho_1, \rho_2, \rho_3, \mu_1, \mu_2, \delta_1, \delta_2$ of D_4. Thus the fifth matrix listed, which corresponds to $\mu_1 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$ is the matrix obtained from the identity by interchanging row 1 with row 2 and row 3 with row 4.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

23. The identity and flipping over on the vertical axis that falls on the vertical line segment of the figure give the only symmetries. The symmetry group is isomorphic to $\mathbb{Z}_2$.

24. As symmetries other than the identity, the figure admits a rotation through 180°, a flip in the vertical line shown, and a flip in the analogous horizontal line (not shown). This group of four elements is isomorphic to the Klein 4-group.

25. If we join endpoints of the line segments, we have a square with the given lines as its diagonals. The symmetries of that square produce all the symmetries of the given figure, so the group of symmetries is isomorphic to D_4.

26. The only symmetries are those obtained by sliding the figure to the left or to the right. We consider the vertical line segments to be one unit apart. For each integer n, we can slide the figure n units to the right if $n > 0$ and $|n|$ units to the left if $n < 0$, leaving the figure alone if $n = 0$. A moment of thought shows that performing the symmetry coresponding to an integer n and then the one corresponding to an integer m yields the symmetry corresponding to $n + m$. We see that the symmetry group is isomorphic to $\mathbb{Z}$.

27. (See the answer in the text.)

28. Replace the final "to" by "onto".

A **permutation** of a set S is a one-to-one map of S onto S.

29. The definition is correct.

30. This one-to-one map of $\mathbb{R}$ onto $\mathbb{R}$ is a permutation.

31. This is not a permutation; it is neither one to one nor onto. Note that $f_2(3) = f_2(-3) = 9$ and $f_2(x) = -1$ has no solution.

32. This one-to-one map of $\mathbb{R}$ onto $\mathbb{R}$ is a permutation.

33. This is not a permutation, it is not a map *onto* $\mathbb{R}$. Note that $f_4(x) \neq -1$ for any $x \in \mathbb{R}$.

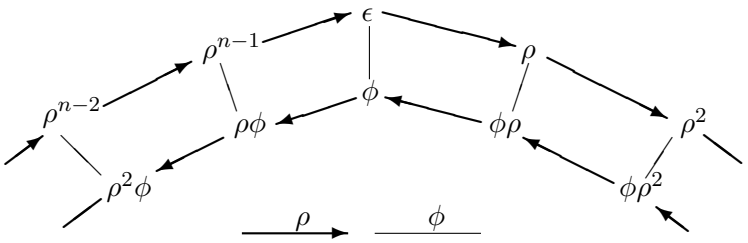
34. This is not a permutation. Note that $f_5(2) = f_5(-1) = 0$, so f_5 is not one to one.

35. T F T T T T F F F T

36. Every proper subgroup of S_3 is abelian, for such a subgroup has order either 1, 2, or 3 by Exercise 18b.

37. Function composition is associative and there is an identity element, so we have a *monoid*.

38. Let ρ denote the rotation through $2\pi/n$ radians and let ϕ denote the reflection (flip) an axis through a vertex that bisects the vertex angle there. The diagram below shows the top part of a Cayley digraph consisting of two concentric n-gons whose $2n$ vertices correspond to the elements of D_n. We let ϵ denote the identity element.



39. If x is a fixed element of G, then mapping each g in G into xg gives a permutation λ_x of G. The map ϕ of G into S_G that carries each x in G into λ_x is then an isomorphism of G with a subgroup of the group S_G.

40. Yes, it is a subgroup.

Closure: If $\sigma(b) = b$ and $\mu(b) = b$, then $(\sigma\mu)(b) = \sigma(\mu(b)) = \sigma(b) = b$.

Identity: The identity carries every element into itself, and hence carries b into b.

Inverses: If $\sigma(b) = b$, then $\sigma^{-1}(b) = b$.

41. No, the set need not be closed under the operation if B has more than one element. Suppose that σ and μ are in the given set, that $b, c \in B$ and $\sigma(b) = c$ but that $\mu(c) \notin B$. Then $(\mu\sigma)(b) = \mu(\sigma(b)) = \mu(c) \notin B$, so $\mu\sigma$ is not in the given set.

42. No, an inverse need not exist. Supppose $A = \mathbb{Z}$ and $B = \mathbb{Z}^+$, and let $\sigma : A \to A$ be defined by $\sigma(n) = n + 1$. Then σ is in the given set, but σ^{-1} is not because $\sigma^{-1}(1) = 0 \notin \mathbb{Z}^+$.

43. Yes, it is a subgroup. Use the proof in Exercise 40, but replace b by B and ( ) by [ ] everywhere.

44. The order of D_n is $2n$ because the regular n-gon can be rotated to n possible positions, and then turned over and rotated to give another n positions. The rotations of the n-gon, without turning it over, clearly form a cyclic subgroup of order n.

45. The group has 24 elements, for any one of the 6 faces can be on top, and for each such face on top, the cube can be rotated in four different positions leaving that face on top. The four such rotations, leaving the top face on top and the bottom face on the bottom, form a cyclic subgroup of order 4. There are two more such rotation groups of order 4, one formed by the rotations leaving the front and back faces in those positions, and one formed by the rotations leaving the side faces in those positions. One exhibits a subgroup of order three by taking hold of a pair of diagonally opposite vertices and rotating through the three possible positions, corresponding to the three edges emanating from each vertex. There are four such diagonally opposite pairs of vertices, giving the desired four groups of order three.

46. Let $n \geq 3$, and let $\rho \in S_n$ be defined by $\rho(1) = 2, \rho(2) = 3, \rho(3) = 1$, and $\rho(m) = m$ for $3 < m \leq n$. Let $\mu \in S_n$ be defined by $\mu(1) = 1, \mu(2) = 3, \mu(3) = 2$, and $\mu(m) = m$ for $3 < m \leq n$. Then $\rho\mu \neq \mu\rho$ so S_n is not commutative. (Note that if $n = 3$, then ρ is our element ρ_1 and μ is our element μ_1 in S_3.)

47. Suppose $\sigma(i) = m \neq i$. Find $\gamma \in S_n$ such that $\gamma(i) = i$ and $\gamma(m) = r$ where $r \neq m$. (Note this is possible because $n \geq 3$.) Then $(\sigma\gamma)(i) = \sigma(\gamma(i)) = \sigma(i) = m$ while $(\gamma\sigma)(i) = \gamma(\sigma(i)) = \gamma(m) = r$, so $\sigma\gamma \neq \gamma\sigma$. Thus $\sigma\gamma = \gamma\sigma$ for all $\gamma \in S_n$ only if σ is the identity permutation.

48. Let c be an element in both $\mathcal{O}_{a,\sigma}$ and $\mathcal{O}_{b,\sigma}$. Then there exist integers r and s such that $\sigma^r(a) = c$ and $\sigma^s(b) = c$. Then $\sigma^{r-s}(a) = \sigma^{-s}(\sigma^r(a)) = \sigma^{-s}(c) = b$. Therefore, for each integer $n \in \mathbb{Z}$, we see that $\sigma^n(b) = \sigma^{n+r-s}(a)$. Hence $\{\sigma^n(b) \mid n \in \mathbb{Z}\} = \{\sigma^n(a) \mid n \in \mathbb{Z}\}$.

49. Let $A = \{a_1, a_2, \cdots a_n\}$. Let $\sigma \in S_A$ be defined by $\sigma(a_i) = a_{i+1}$ for $1 \leq i < n$ and $\sigma(a_n) = a_1$. (Note that σ essentially performs a rotation if the elements of A are spaced evenly about a circle.) It is clear that σ^n is the identity permutation and $|\langle\sigma\rangle| = n = |A|$. We let $H = \langle\sigma\rangle$. Let a_i and a_j be given; suppose $i < j$. Then $\sigma^{j-i}(a_i) = a_j$ and $\sigma^{i-j}(a_j) = a_i$, so H is transitive on A.

50. Let $\langle\sigma\rangle$ be transitive on A and let $a \in A$. Then $\{\sigma^n(a) \mid n \in \mathbb{Z}\}$ must include all elements of A, that is, $\mathcal{O}_{a,\sigma} = A$.

Conversely, suppose that $\mathcal{O}_{a,\sigma} = A$ for some $a \in A$. Then $\{\sigma^n(a) \mid n \in \mathbb{Z}\} = A$. Let $b, c \in A$ and let $b = \sigma^r(a)$ and $c = \sigma^s(a)$. Then $\sigma^{s-r}(b) = \sigma^s(\sigma^{-r}(b)) = \sigma^s(a) = c$, showing that $\langle\sigma\rangle$ is transitive on A.

51. **a.** The person would see all possible products $a *' b$ and all instances of the associative property for $*'$ in G'.

b. *Associativity:* Let $a, b, c \in G'$. Then $a *' (b *' c) = a *' (c * b) = (c * b) * a = c * (b * a) = (b * a) *' c = (a *' b) *' c$ where we used the fact that G is a group and the definition of $*'$.

Identity: We have $e *' a = a * e = a$ and $a *' e = e * a = a$ for all $a \in G'$.

Inverses: Let $a \in G'$ and let a^{-1} be the inverse of a in G. Then $a^{-1} *' a = a * a^{-1} = e = a^{-1} * a = a *' a^{-1}$, so a^{-1} is also the inverse of a in G'.

52. To start, we show that ρ_a is a permutation of G. If $\rho_a(x) = \rho_a(y)$, then $xa = ya$ and $x = y$ by group cancellation, so ρ_a is one to one. Because $\rho_a(xa^{-1}) = xa^{-1}a = x$, we see that ρ_a maps G onto G. Thus ρ_a is a permutation of the set G. Let $G'' = \{\rho_a \mid a \in G\}$.

For $a, b \in G$, we have $(\rho_a\rho_b)(x) = \rho_a(\rho_b(x)) = \rho_a(xb) = xba = \rho_{ba}(x)$, showing that G'' is closed under permutation multipliction. Because ρ_e is the identity permutation and because $\rho_{a^{-1}}\rho_a = \rho_e$, we see that G'' is a subgroup of the group S_G of all permutations of G.